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[Submitted on 16 Dec 2021 (v1), last revised 22 Nov 2022 (this version, v3)]

Trading with the Momentum Transformer: An Intelligent and Interpretable Architecture

Kieran Wood, Sven Giegerich, Stephen Roberts, Stefan Zohren

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We introduce the Momentum Transformer, an attention-based deep-learning architecture, which outperforms benchmark time-series momentum and mean-reversion trading strategies. Unlike state-of-the-art Long Short-Term Memory (LSTM) architectures, which are sequential in nature and tailored to local processing, an attention mechanism provides our architecture with a

direct connection to all previous time-steps. Our architecture, an attention-LSTM hybrid, enables us to learn longer-term dependencies, improves performance when considering returns net of transaction costs and naturally adapts to new market regimes, such as during the SARS-CoV-2 crisis. Via the introduction of multiple attention heads, we can capture concurrent regimes, or temporal dynamics, which are occurring at different timescales. The Momentum Transformer is inherently interpretable, providing us with greater insights into our deep-learning momentum trading strategy, including the importance of different factors over time and the past time-steps which are of the greatest significance to the model.

Comments: included motivation for attention mechanism and additional architecture details
Subjects: **Machine Learning (cs.LG)**; Trading and Market Microstructure (q-fin.TR); Machine Learning (stat.ML)
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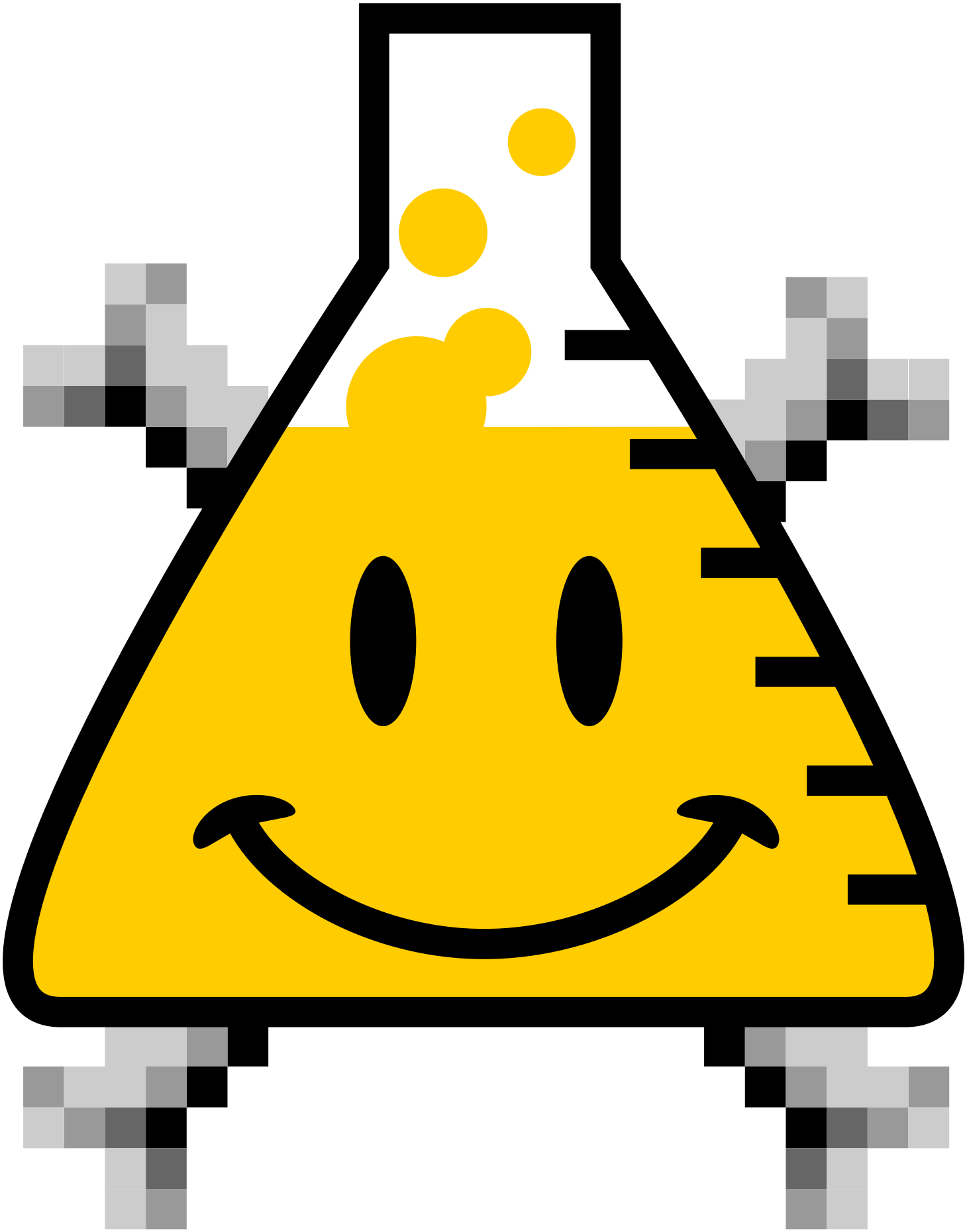

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